Capstone Checkpoint 4.1 - ESTABLISHED BASE - validation-set RAGAS evaluation

Oracle, vector, and hybrid experiment comparison. Generated 2026-09-13 12:53 EDT.

7	2	175	25	0
Included runs	Configurations	Question-runs	Questions	Legacy runs excluded

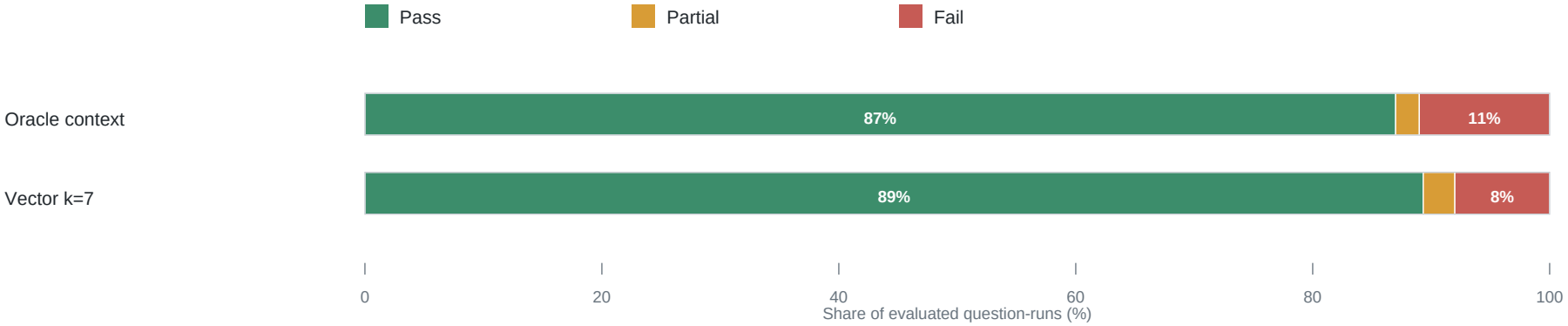
Summary

The highest observed mean quality score was 0.907 for Vector k=7 across 3 run(s). Quality assigns 1 to pass, 0.5 to partial, and 0 to fail.

The highest observed mean designated-source recall was 97.3% for Vector k=7. Article recall is recomputed from the required and retrieved article names in every CSV.

The highest task-family attention score was 40.0 for Exhaustive list or time filter under Vector k=7. Attention is 100 x (1 - grouped quality), so higher values identify larger observed improvement opportunities.

Answer outcomes

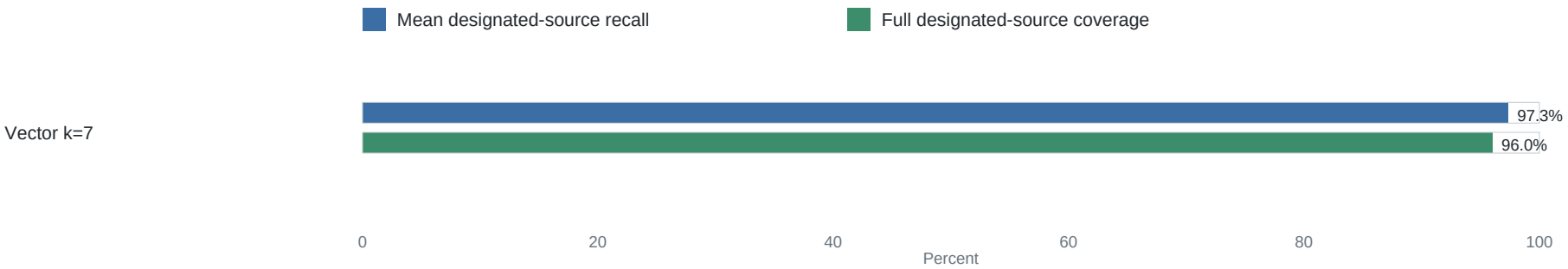


Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
Oracle context	4	100	87.0%	2.0%	11.0%	0.880	0.032	n/a	n/a	58.8k
Vector k=7	3	75	89.3%	2.7%	8.0%	0.907	0.019	97.3%	96.0%	130.9k

Percentages aggregate all question-runs within a configuration. Run SD is the standard deviation of mean quality across repeated runs.

Retrieval diagnostics

Designated-source coverage



The analyzer recomputes recall after removing only final .txt and .html suffixes. This corrects historical .html.txt versus .html mismatches. Designated-source recall is diagnostic: alternative retrieved articles may still contain sufficient evidence.

Outcome and source-coverage combinations

Configuration	Pass, full	Pass, incomplete	Non-pass, full	Non-pass, missing
Vector k=7	67	0	5	3

A non-pass result with full designated-source recall points toward answer synthesis, context noise, table interpretation, or judge behavior. A non-pass result with a missing designated source is consistent with a retrieval recall problem, but manual review is still required.

Per-question comparison

Cells show the mean quality score for each question and configuration across repeated runs.

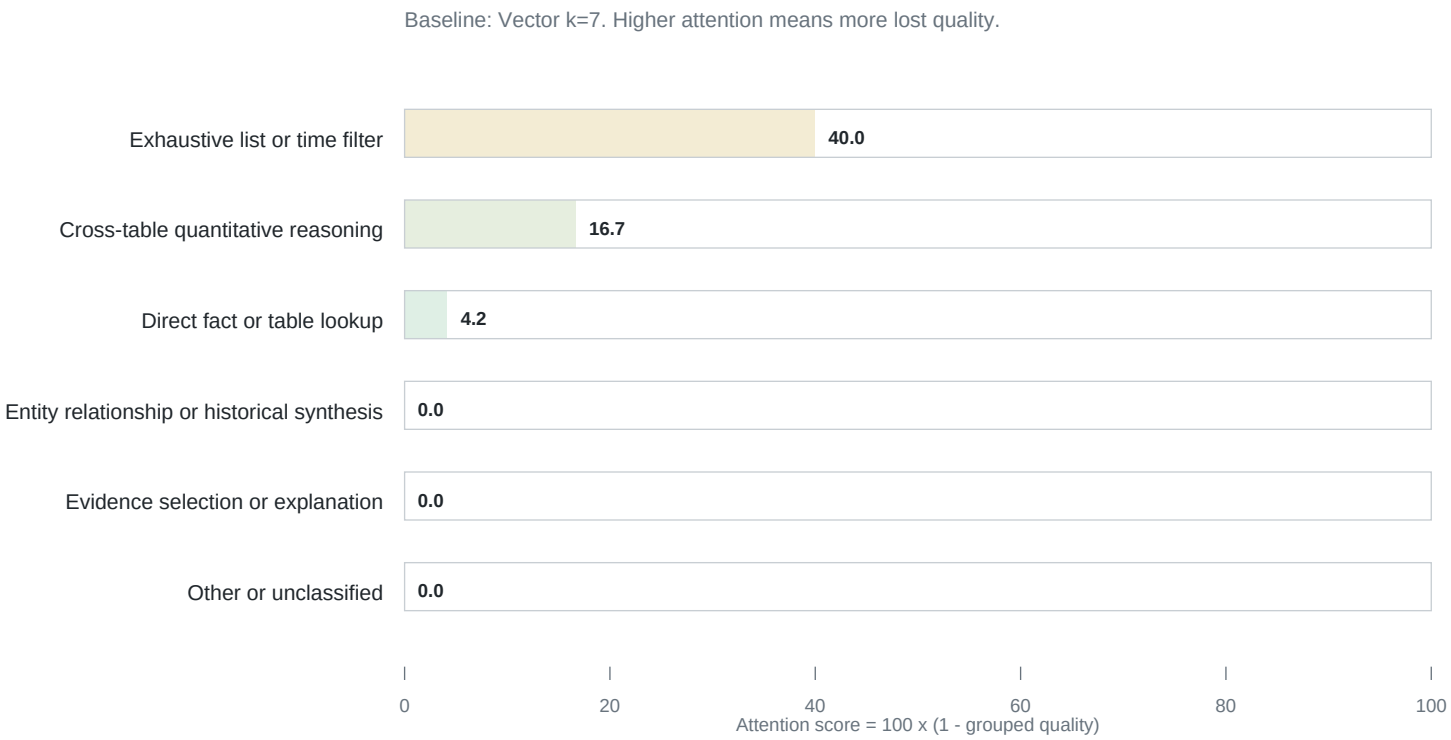
	Oracle context	Vector k=7
VAL01	**1.00**	**1.00**
VAL02	**0.00**	**0.00**
VAL03	**1.00**	**1.00**
VAL04	**1.00**	**1.00**
VAL05	**0.12**	**1.00**
VAL06	**0.75**	**1.00**
VAL07	**0.25**	**0.00**
VAL08	**1.00**	**1.00**
VAL09	**1.00**	**1.00**
VAL10	**1.00**	**0.83**
VAL11	**1.00**	**1.00**
VAL12	**1.00**	**1.00**
VAL13	**1.00**	**1.00**
VAL14	**0.88**	**0.83**
VAL15	**1.00**	**1.00**
VAL16	**1.00**	**1.00**
VAL17	**1.00**	**1.00**
VAL18	**1.00**	**1.00**
VAL19	**1.00**	**1.00**
VAL20	**1.00**	**1.00**
VAL21	**1.00**	**1.00**
VAL22	**1.00**	**1.00**
VAL23	**1.00**	**1.00**
VAL24	**1.00**	**1.00**
VAL25	**1.00**	**1.00**

Fail (0) Partial (0.5) Pass (1)

Question	Oracle context	Vector k=7
VAL01	1.000	1.000
VAL02	0.000	0.000
VAL03	1.000	1.000
VAL04	1.000	1.000
VAL05	0.125	1.000
VAL06	0.750	1.000
VAL07	0.250	0.000
VAL08	1.000	1.000
VAL09	1.000	1.000
VAL10	1.000	0.833
VAL11	1.000	1.000
VAL12	1.000	1.000
VAL13	1.000	1.000
VAL14	0.875	0.833
VAL15	1.000	1.000
VAL16	1.000	1.000
VAL17	1.000	1.000
VAL18	1.000	1.000
VAL19	1.000	1.000
VAL20	1.000	1.000
VAL21	1.000	1.000
VAL22	1.000	1.000
VAL23	1.000	1.000
VAL24	1.000	1.000
VAL25	1.000	1.000

Performance by task type

The 18 original category labels are consolidated into five task families so each group contains multiple questions. The attention score ranks observed improvement opportunities for the strongest retrieval configuration; it is not a statistical significance test.

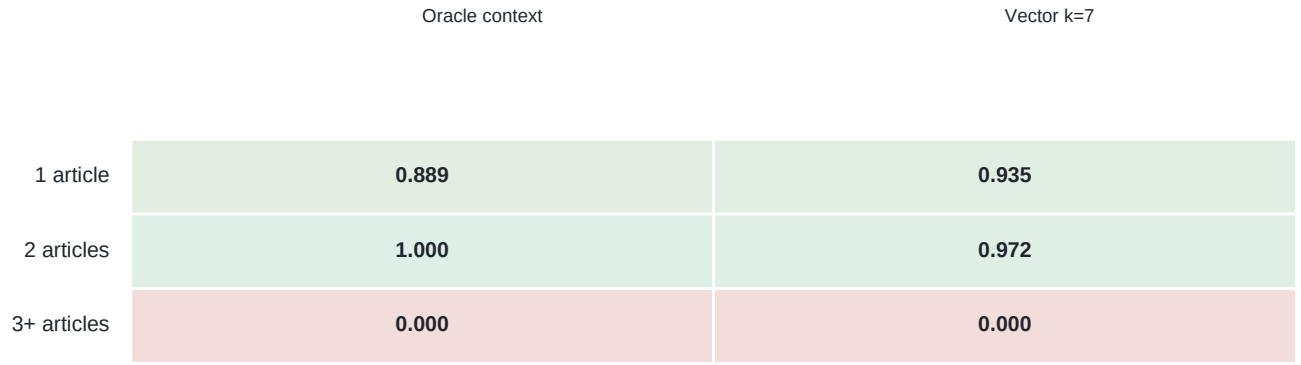


Task family	Questions	Baseline quality	Attention	Fail	Oracle quality	Oracle gap	Article recall
Exhaustive list or time filter	5	0.600	40.0	40.0%	0.475	-0.125	86.7%
Cross-table quantitative reasoning	1	0.833	16.7	0.0%	1.000	+0.167	100.0%
Direct fact or table lookup	4	0.958	4.2	0.0%	0.969	+0.010	100.0%
Entity relationship or historical synthesis	1	1.000	0.0	0.0%	1.000	+0.000	100.0%
Evidence selection or explanation	10	1.000	0.0	0.0%	0.975	-0.025	100.0%
Other or unclassified	4	1.000	0.0	0.0%	1.000	+0.000	100.0%

Baseline quality and failure rate use Vector k=7. Oracle gap equals oracle quality minus baseline quality: a large positive gap is consistent with a retrieval or context-selection opportunity, while low oracle quality indicates that the difficulty persists even when the designated articles are supplied.

Performance by source scope

Source scope groups questions by the number of designated articles. It tests whether the number of sources, rather than the reasoning operation, explains performance.

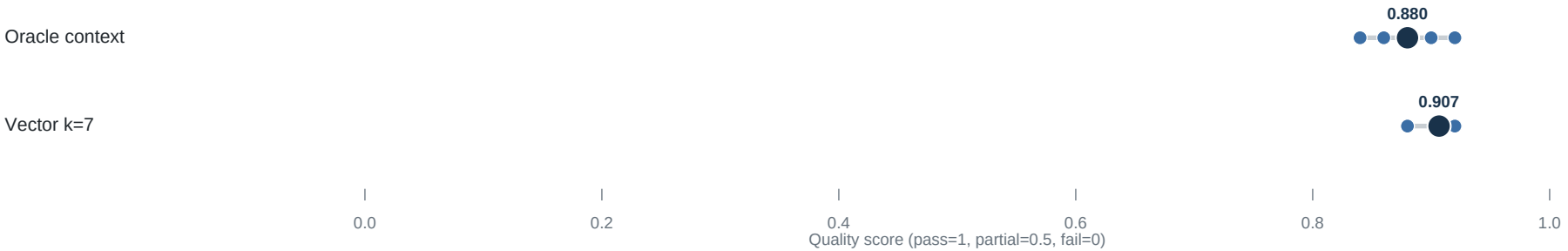


Source scope	Configuration	Questions	Question-runs	Pass	Partial	Fail	Quality	Article recall
1 article	Oracle context	18	72	87.5%	2.8%	9.7%	0.889	n/a
2 articles	Oracle context	6	24	100.0%	0.0%	0.0%	1.000	n/a
3+ articles	Oracle context	1	4	0.0%	0.0%	100.0%	0.000	n/a
1 article	Vector k=7	18	54	92.6%	1.9%	5.6%	0.935	100.0%
2 articles	Vector k=7	6	18	94.4%	5.6%	0.0%	0.972	100.0%
3+ articles	Vector k=7	1	3	0.0%	0.0%	100.0%	0.000	33.3%

Question-runs are repeated measurements of the listed unique questions. Source-scope groups remain small and mix different task families, so this view should be used as a diagnostic rather than a general performance estimate.

Run-to-run variability

Each dot is one run; the larger dot is the configuration mean.



Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
epic_mccarthy	Oracle context	25	22	1	2	0.900	n/a	n/a
confident_ullman	Oracle context	25	23	0	2	0.920	n/a	n/a
quirky_kernighan	Oracle context	25	21	0	4	0.840	n/a	n/a
bold_rivest	Oracle context	25	21	1	3	0.860	n/a	n/a
focused_knuth	Vector k=7	25	23	0	2	0.920	97.3%	96.0%
quizzical_goldwasser	Vector k=7	25	23	0	2	0.920	97.3%	96.0%
romantic_brin	Vector k=7	25	21	2	2	0.880	97.3%	96.0%

Methodology and limitations

Scoring

Pass is assigned 1, partial is assigned 0.5, and fail is assigned 0. The quality score is the arithmetic mean across the applicable question-runs. Pass, partial, and fail percentages are also reported separately so the weighted score does not hide the underlying verdicts.

Task-family attention

The attention score is 100 multiplied by one minus grouped quality. A score of 66 means that the group lost 66% of the available quality points under the selected retrieval baseline. It is a prioritization aid, not an estimate of population-wide failure risk. Task-family and source-scope results must be read with their unique-question counts because the evaluation set contains only 19 questions.

Configuration grouping

Runs are grouped by evaluation mode, retriever kind and label, Top K, hybrid weights, answer model, and judge model. RAGAS-generated filenames identify individual runs but do not define configurations.

Framework compatibility

Legacy oracle runs stored under pass_fail_only directories are excluded by default because they could not produce partial verdicts. Use --include-binary-oracle only for a labeled historical appendix, not for a direct three-level score comparison.

Interpretation limits

An LLM produced the answers and another LLM applied the rubric, so repeated runs can vary. Article recall measures coverage of designated ground-truth sources, not every possible supporting source. Full article recall does not prove that the model used the evidence correctly, and missing a designated source does not prove the context was insufficient when another article contains equivalent evidence. Differences between configurations should be interpreted together with run counts and per-question behavior.

Included experiment files

Run file
bold_rivest.csv
confident_ullman.csv
epic_mccarthy.csv
focused_knuth.csv
quirky_kernighan.csv
quizzical_goldwasser.csv
romantic_brin.csv

Warnings

Task-family analysis left these categories unclassified: Cross-article comparison, Diagnostic reasoning and limitations, Inference limitation, People connection: shared work and roles